

Deep learning versus ophthalmologists for screening for glaucoma on fundus examination: A systematic review and meta-analysis

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Abstract

Background: In this systematic review and meta-analysis, we aimed to compare deep learning versus ophthalmologists in glaucoma diagnosis on fundus examinations.

Method: PubMed, Cochrane, Embase, ClinicalTrials.gov and ScienceDirect databases were searched for studies reporting a comparison between the glaucoma diagnosis performance of deep learning and ophthalmologists on fundus examinations on the same datasets, until 10 December 2020. Studies had to report an area under the receiver operating characteristics (AUC) with SD or enough data to generate one.

Results: We included six studies in our meta-analysis. There was no difference in AUC between ophthalmologists (AUC = 82.0, 95% confidence intervals [CI] 65.4–98.6) and deep learning (97.0, 89.4–104.5). There was also no difference using several pessimistic and optimistic variants of our meta-analysis: the best (82.2, 60.0–104.3) or worst (77.7, 53.1–102.3) ophthalmologists versus the



best (97.1, 89.5–104.7) or worst (97.1, 88.5–105.6) deep learning of each study. We did not retrieve any factors influencing those results.

Conclusion: Deep learning had similar performance compared to ophthalmologists in glaucoma diagnosis from fundus examinations. Further studies should evaluate deep learning in clinical situations.

KEYWORDS

artificial intelligence, deep learning, glaucoma, machine learning, screening

1 | INTRODUCTION

Glaucoma is the leading cause of irreversible blindness, currently affecting 80 million people worldwide and 112 million in 2040.¹ Glaucoma is a chronic neurodegenerative disease of the eye,² asymptomatic in its early stage—50% of patients being unaware of the disease.³ Early treatment of glaucoma can decrease the rate of blindness by 50% 20 years later.⁴ Therefore, screening and early diagnosis of glaucoma is essential for the proper management of the disease.⁵ The cup-to-disc ratio of the optic nerve seen in fundus examinations, which measures the vertical diameter of the optic cup to that of the disc is one of the most commonly adopted screening tests for glaucoma.⁶ However, fundus examination by an ophthalmologist is labour-intensive with learning curve and time-consuming. In addition, it's a subjective test and the efficiency is linked to the experience of the observer. An international survey in 2010 showed that despite over 200 000 ophthalmologists worldwide, there is a significant shortfall in developing countries.⁷ Moreover, the ageing of population in developed countries is more increasing than number of trained practitioners.⁷ To meet this widening gap between need and supply, there is a need for innovative support to improve diagnosis capacity of healthcare systems, and to remedy the specialists shortage. Artificial intelligence (AI) has been successfully applied in image-based ophthalmic diagnosis such as diabetic retinopathy.^{8,9} AI is the development of computer systems able to perform tasks by mimicking human intelligence. Machine learning is a subfield of AI, which enables computers to improve with experience. Deep learning is a subset of machine learning inspired by the human brain with thousands of neurons capable of performing complex tasks.^{10,11}

In this systematic review, we sought to evaluate deep learning as a tool to help the ophthalmologists in the screening of glaucoma. Therefore, we have conducted a meta-analysis to compare the performance of deep learning versus health-care professionals in glaucoma diagnosis on fundus examinations.

2 | METHODS

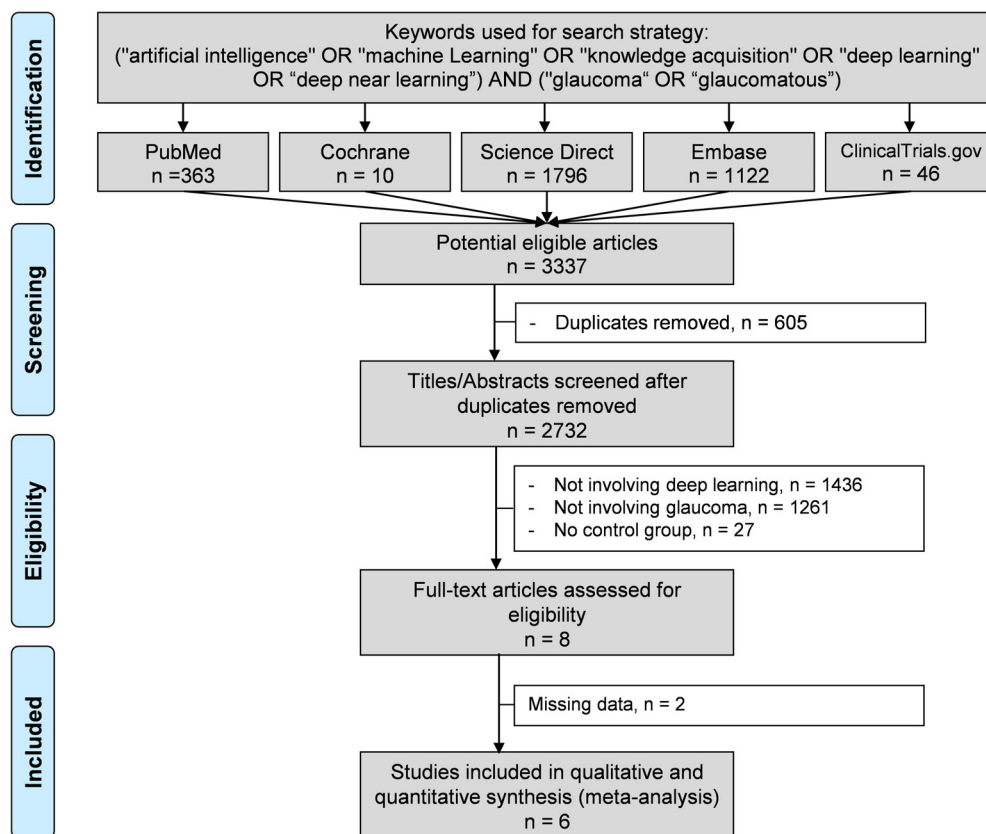
2.1 | Literature search

We reviewed all studies reporting a comparison between the glaucoma diagnosis performance of deep learning and ophthalmologists on fundus examinations. The PubMed, Cochrane Library, Science Direct, Embase and ClinicalTrials.gov databases were searched on 10 December 2020, with the following keywords: “Deep learning” or “machine learning” or “AI” and “glaucoma” (details of search strategy is presented in supplementary material Appendix 1). The search was not limited to specific years and no language restrictions were applied. To be included, articles needed to assess the glaucoma diagnosis performance of deep learning in comparison to ophthalmologists on the same fundus database. Articles needed to report the area under the curve (AUC) value with its SD for both deep learning and ophthalmologists' performances, or at least four sensitivity and specificity points for each group to generate an AUC. The AUC curve is the most common statistical representation demonstrating the ability to assess the performance of a binary classifier across varying discrimination thresholds.¹² Results range from 0.5 to 1. The closer the result is to one, the better the performance. The search strategy is presented in Figure 1. Two authors (Mathieu Buisson and Valentin Navel) conducted all literature searches, collated and reviewed the abstracts and based on the selection criteria, decided the suitability of the articles for inclusion. A third author (Frédéric Dutheil) was asked to review the articles where consensus on suitability was debated. All authors then reviewed the eligible articles. We followed the PRISMA guidelines (Appendix 2).

2.2 | Data collection

The data collected included first author's name, publication year, study design (case-control, cross-sectional or cohort study), aims and outcomes of included articles,

FIGURE 1 Search strategy



characteristics of ophthalmologists (number, experience, age and sex), characteristics of testing sets (inclusion and exclusion criteria, population size, age and sex), the deep learning technique, the validation technique of the deep learning system, the reference/gold standard for glaucoma in the population, the AUC value with its SD, the sensitivity and specificity with their SD and the contingency tables.

2.3 | Quality of assessment

Each study was assessed by two scales: QUADAS-2 and an adapted version of the Newcastle-Ottawa Scale (NOS). The QUADAS-2 is a tool developed to assess potential bias in systematic reviews of diagnostic accuracy tests. We attributed one point per item, then converted into percentage to give a quality score for each included study, the maximum score was 7.¹³ We also used an adapted version of the NOS developed by Murtagh et al. in a systematic review comparing the performance of deep learning on ocular coherence tomography and fundal photography, in terms of diagnostic accuracy in the screening and diagnosis of glaucoma.¹⁴ The maximum score was six, studies which scored four points or greater were deemed to be of a higher methodological standard.^{14,15} The methodological quality of the studies was evaluated by two authors

(Mathieu Buisson and Valentin Navel). Disagreements were addressed by obtaining a consensus with a third author (Frédéric Dutheil) (Figure 2 and Appendix 3).

2.4 | Statistical considerations

Statistical analyses were conducted using Stata software (version 16, StataCorp, College Station, US).^{16–20} Main characteristics were synthesized for each study population and reported as the mean $\pm$ SD for continuous variables and the number (%) for categorical variables. We conducted random effects meta-analyses (DerSimonian and Laird approach) on the performance on glaucoma diagnosis using AUC values.²¹ A ROC curve is generated by plotting (1-specificity) on the x-axis and sensitivity on the y-axis. For studies that reported only sensitivity and specificity, we calculated an overall summary AUC value by estimating a Hierarchical summary ROC curve using the Metandi and Midas packages.²² We stratified meta-analyses of AUC values on glaucoma diagnosis from deep learning and from ophthalmologists. Then, we conducted sensitivity analyses using only the best and/or worst performance for deep learning and for ophthalmologists within each included study—in order to assess non-inferiority of deep learning even using the most unfavourable data that is, best performances for ophthalmologists, and worst



FIGURE 2 Quality of articles–QUADAS2

	Selection bias				Index test			Reference standard			Flow and Timing			
	Consecutive or random sample of patients enrolled	Case-control design avoided	Avoidance of inappropriate exclusions	Included patients matching the review question	Blind to the reference standard	Pre-specified threshold	Index test matching the review question	Correctly classifying the disease	Blind to the results of the index test	Definition by the gold standard matching the question	Appropriate interval between index test and reference	All patient having a reference standard	All patients having the same reference	All patients included in the analysis
Al Aswad 2019	+	-	+	+	+	?	+	+	+	+	+	+	+	+
Jammal 2019	+	-	+	+	+	?	+	+	+	+	-	+	+	+
Liu 2018	+	-	+	+	+	?	+	+	+	+	+	+	+	+
Phene 2019	+	-	+	+	+	?	+	+	+	+	+	+	+	+
Seo 2019	+	-	+	+	+	?	+	+	+	+	-	+	+	+
Shibata 2018	+	-	+	+	+	?	+	+	+	+	+	+	+	+

performances for deep learning. We evaluated heterogeneity in the study results by examining forest plots, confidence intervals (CI) and I-squared (I^2). Formal tests for homogeneity based on the I^2 statistic are the most common metric for measuring the magnitude of between-study heterogeneity and are easily interpretable. I^2 values range from 0% to 100%, and are considered low for <25%, modest for 25%–50%, and high for >50%. We further searched for potential publication bias using funnel plots of all aforementioned meta-analyses, and we verified the strength of our results by conducting further meta-analyses after exclusion of studies that were not evenly distributed around the base of the funnel. We performed meta-regressions to compare performances of deep learning and ophthalmologists. When possible (sufficient sample size), meta-regressions were also proposed to study the relationship between AUC and characteristics of population such as sociodemographic or characteristics of glaucoma. Results were expressed as regression coefficients and 95% CI. p values less than 0.05 were considered statistically significant.

3 | RESULTS

An initial search produced a possible for 3331 articles (Figure 1). Removal of duplicates and use of the selection

criteria reduced the number of articles reporting a comparison between the glaucoma diagnosis performance of deep learning and ophthalmologists on fundus examinations to eight articles that were included in the systematic review (Table 1). All articles were written in English. Two studies were excluded because of the lack of data.^{23,24} Therefore, we performed the meta-analysis on six studies.^{25–30}

3.1 | Quality of articles

The six included studies performed well in quality assessment, with scores ranging from 78.5%^(28,29) to 85.7%^(25–27,30) using the QUADAS-2 tool, and from 56.7%^(25–29) to 83.3%⁽³⁰⁾ using the adapted NOS scale (Figure 2 and Appendix 3).

3.2 | Study designs and objectives

All studies included were observational studies. Three studies were conducted in the United States of America,^{25,27–29} one in Japan³⁰ and one in Australia.²⁶ All included studies had for objective to compare screening performance for glaucoma on fundus examination between deep learning system and ophthalmologists. Fundus images were extracted from one^{25,28–30} or multiple^{26,27} sources.



TABLE 1 Descriptive characteristics of studies

Study	Country	Design	Characteristics of testing sets			Deep learning		Ophthalmologists			
			n (%GON)	Age, mean $\pm$ SD	Sex, %Men	Architecture	Training set (n images)	Validation method	n	Experience	Age Sex Gold standard
Al-Aswad 2019 ²⁵	USA	Observational, single-site	110 (50)	All: 58 $\pm$ 11	—	Pegasus	136 146	—	6	Residents to Glaucoma specialists	— — ORIGA
Liu 2018 ²⁶	Australia	Observational, three sites	80 (50)	GON: 60.9 $\pm$ 12.8 Control: 55.6 $\pm$ 12.2	57.7 48.3	ResNet50	3200	—	18	General ophthalmologists	— — Glaucoma Specialists RIM-ONE
Phene 2019 ²⁷	USA	Observational, multicentric	411 (42.6)	All: 57 $\pm$?	46.8	Inception V3	86 618	—	7	Ophthalmologists, glaucoma specialists	— — Glaucoma specialists
Shibata 2018 ³⁰	Japan	Observational, single-site	110 (55.5)	GON: 68.7 $\pm$ 7.9 Control: 66.9 $\pm$ 10.3	36 56	Resnet 18 VGG-16	3132	Cross validation	3	Residents	— — Glaucoma specialists
Seo 2018 ²⁸	USA	Observational, single-site	186 (43.5)	GON: 63.4 $\pm$ 15.8 Control: 57.2 $\pm$ 13.9	55.6 49.5	Pegasus	N/A	—	2	Glaucoma specialists	— — OCT and Repeatable visual fields
Jammal 2019 ²⁹	USA	Observational, single-site	490 (42.9)	GON: 63.4 $\pm$ 13.1 Control: 58.4 $\pm$ 14.3	53.8 42.5	ResNet34	26 256	—	2	Glaucoma specialists	— — Duke glaucoma repository
Li 2020 ²³	China	Observational, two sites	3481 (41.4)	—	—	ResNet101	23 104	Cross validation	2	Glaucoma specialists	— — Ophthalmologists
Rogers 2019 ²⁴	United Kingdom	Observational, single-site	94 (51.1)	—	—	Pegasus	125 934	—	243	Ophthalmologists	— — Glaucoma specialists

Abbreviations: GON, glaucomatous optic neuropathy; N, number of fundus in the testing set; n°, number of; SD, standard deviation.

3.3 | Inclusion and exclusion criteria

Inclusion criteria of glaucoma fundus images were glaucomatous appearance of the optic nerve according to the recommendations of the Japan Glaucoma Society Guidelines for Glaucoma³⁰ or The American Academy of Ophthalmology's Primary Open-Angle Glaucoma Preferred Practice Patterns.²⁷ Exclusion criteria were other optic nerve or retinal pathologies that could be a confounding factor,^{27,30} or poor quality images.^{26,27,30} Some studies did not report inclusion nor exclusion criteria for the glaucoma and normal images.^{25,28,29}

3.4 | Population studied

3.4.1 | Sample size

We included a total of 1392 eyes: 684 with glaucoma and 708 without glaucoma. Sample sizes ranged from 80²⁶ to 490²⁹ eyes. We could not report the number of patients because the studies included in our meta-analysis did not share the same criteria for duplicates and inclusion of one or both eyes of the patients.

3.4.2 | Age

Mean age was 65.6 years old (95% CI 54.6–76.5) in the glaucoma group, ranging from 60.2 ± 12.1^{30myopic}–68.7 ± 7.9,^{(30)normal} and 60.5 years old (95% CI 48.4–72.6) in the control group, ranging from 42.1 ± 16.6^{30myopic}–66.9 ± 10.3 years old^{30normal}. One study did not report SD.²⁷

3.4.3 | Gender

In the glaucoma group, there was 52% men (95% CI 45%–59%), ranging from 39.3%³⁰–57.7%.²⁶ In the control group, there was 45% men (95% CI 40%–49%), ranging from 42.5%²⁹–49.5%.²⁸ One study reported gender without discriminating glaucoma or control group;²⁷ and one study did not report gender.²⁵

3.4.4 | Ethnicity

The percentage of Caucasians ranged from 53.8%²⁹ to 93.5%²⁶ in the glaucoma group and from 62.1%²⁹ to 90.1%²⁶ in the control group, African-Americans ranged from 16.0%²⁸ to 46.2%²⁹ in the glaucoma group and from 11.4%²⁸ to 37.9%²⁹ in the control group. Other or

unknown ethnicities represented less than 10.0% each in both groups.

3.4.5 | Training datasets

Our meta-analysis included a total of 255 352 training fundus images, ranging from 3132 fundus³⁰ to 136 146 fundus.²⁵ One study²⁸ did not report the training dataset.

3.4.6 | Others

Only one study³⁰ reported the best corrected visual acuity and the intra-ocular pressure, another study²⁸ reported the number of glaucoma eye drops and the mean deviation from Humphrey visual fields. The two aforementioned studies reported the optic disc excavation (vertical cup/disc ratio).

3.5 | Glaucoma

In our meta-analysis, the average proportion of glaucoma was 48% (95% CI 43–54) of the total samples, ranging from 42.6%²⁷ to 60.1%.²⁶ Most studies did not specify the type of glaucoma,^{27–30} the two other studies included only²⁶ or mostly (69%)²⁵ open-angle glaucoma. Fundus images were collected from ophthalmic clinical centers^{26,27,29,30} or datasets^{25–27} such as the publicly available online retinal fundus image database for glaucoma analysis and research (ORIGA)³¹ EyePacs,³² Inoveon, Age-Related Eye Disease study³³ or United Kingdom Biobank. The diagnosis of glaucoma used as the gold standard was extracted from the online datasets^{25–27} or established by the clinicians in each center with the complete medical record.^{26,27,29,30} One study used a combination of optical coherence tomography and repeatable abnormal visual field as the gold standard for glaucoma.²⁸

3.6 | Ophthalmologists

In each included study, there were two^{28,29} to 18²⁶ ophthalmologists who graded fundus photographs independently, for a total of 38 ophthalmologists. They were masked to all patient clinical information and medical records. They had varying levels of glaucoma expertise, ranging from residents in training^{25,30} to general ophthalmologists^{25–27} and glaucoma specialists.^{25,28,29} No other information was given regarding ophthalmologists, such as age, sex, or years of experience.

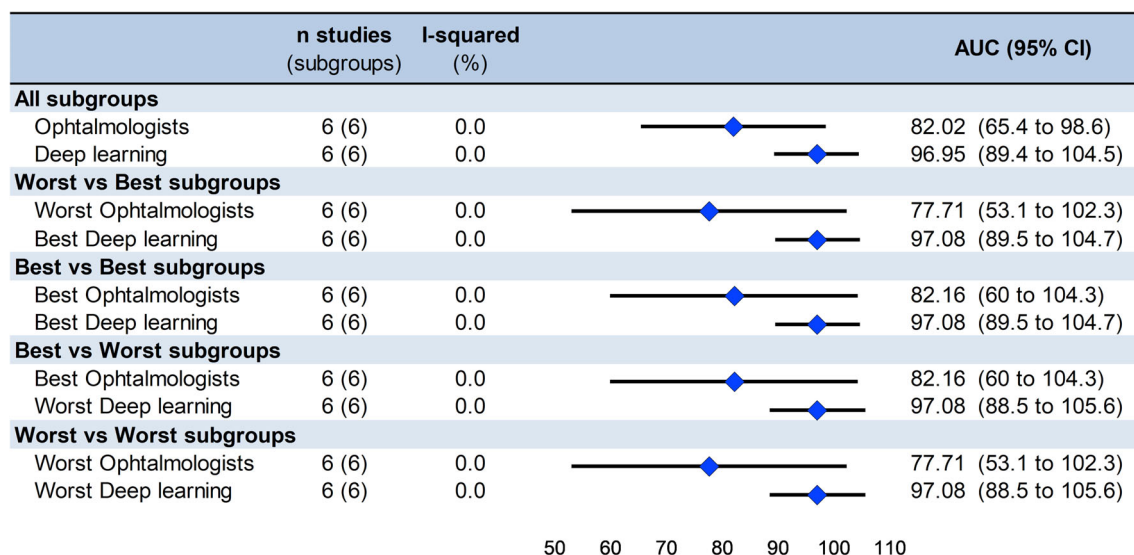


FIGURE 3 Meta-analysis synthesis

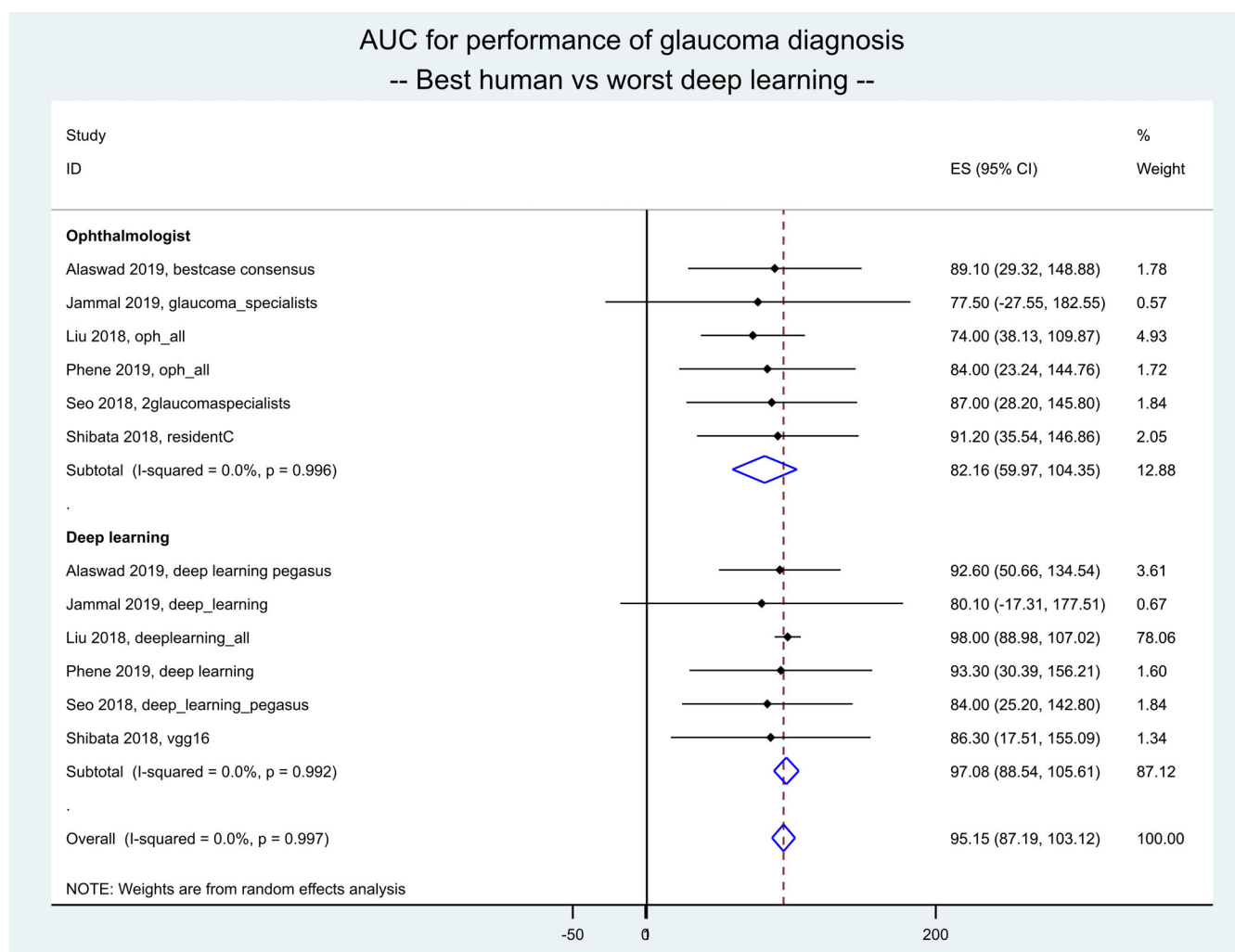


FIGURE 4 Best ophthalmologists versus worst deep learning

3.7 | Deep learning

The studies included in the meta-analysis used different architectures of deep learning: Pegasus,^{25,28} ResNet,^{26,29,30} VGG-16³⁰ and Inception-V3.²⁷ The adjective “deep” refers to the multiple layers between the input data and the outputs. The datasets are divided into three parts in order to train and test the algorithm: training, validation, and testing. The training dataset is a set of examples in which the algorithm learns. The validation one is a set of examples used to tune the deep learning. The testing one is a set of data independent from the training and the validation sets.³⁴ In our meta-analysis, the testing sets were the fundus photographs graded by the ophthalmologists.

3.8 | Meta-analysis on the performance of glaucoma diagnosis

We conducted a meta-analysis with all the AUC in both subgroups, ophthalmologists and deep learning, we reported a respective AUC of 82.0 (95% CI 65.4–98.6) and 97.0 (89.4–104.5). Several pessimistic and optimistic variants of our meta-analysis were conducted: the best (AUC 82.2, 95% CI 60.0–104.3) or worst (77.7, 53.1–102.3) ophthalmologists versus the best (97.1, 89.5–104.7) or worst (97.1, 88.5–105.6) deep learning of each study, that is, best ophthalmologists versus worst deep learning, best ophthalmologists versus best deep learning, worst ophthalmologists versus worst deep learning, and worst ophthalmologists versus best deep learning. All aforementioned meta-analysis had a low degree of heterogeneity (Figures 3 and 4, Appendix 4 to 7).

3.9 | Meta-regressions and sensitivity analysis

The performance of deep learning did not differ from those of ophthalmologists regardless of the model used: all ophthalmologists data versus all deep learning data

(Coefficient 14.9, 95% CI –4.5–34.4), best ophthalmologists versus worst deep learning (14.9, –12.1–42.0), best ophthalmologists versus best deep learning (14.9, –11.7–41.6), worst ophthalmologists versus worst deep learning (19.4, –10.2–48.9), worst ophthalmologists versus best deep learning (19.4, –9.9–48.6). Meta-regressions on age and gender were not statistically significant either (Figure 5). Funnel plots of meta-analyses analysing for potential publication bias are presented in Appendix 8. All results were included in our funnel plot analysis, showing homogeneity between studies (all I² were at 0%).

4 | DISCUSSION

Glaucoma is the leading cause of irreversible blindness, requiring early diagnosis and treatment to decrease the rate of irreversible complications that is, progressive loss of retinal ganglion cells and visual field. We highlighted that deep learning in glaucoma diagnosis was comparable to trained ophthalmologists in the assessment of the optic nerve seen in fundus examinations, even in a pessimist model (worst deep learning versus best ophthalmologists).

4.1 | The performance of deep learning

The main finding of our study was that the performance of the deep learning in glaucoma diagnosis from fundus examinations was comparable to the current model based on the assessment of fundus by trained ophthalmologists. This high-level performance of the deep learning is confirmed by the comparison between the worst deep learning and the best ophthalmologists, which highlighted a similar performance. This finding is consistent with the literature. Many articles compared deep learning with state-of-the-art models on various datasets and achieved a robust performance.^{35–42} It has been shown that deep learning are not only effective in fundus photographs for the diagnosis of glaucoma but also in Optical Coherence

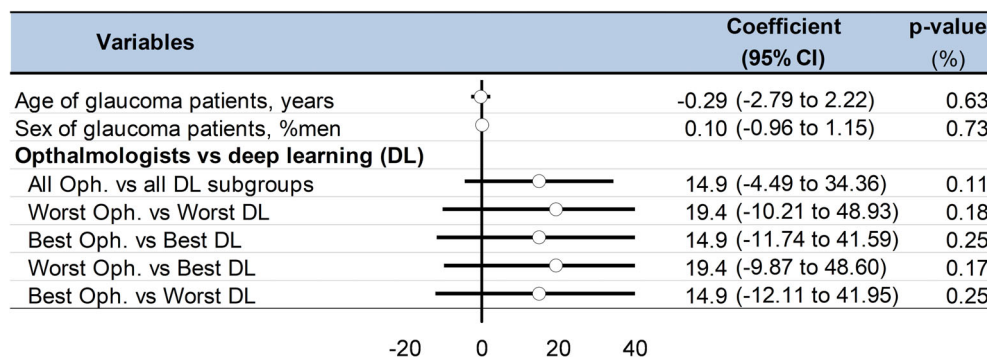


FIGURE 5 Meta-regressions



Tomography^{43–48} and visual fields.^{49–52} AI has been showed effective in other ophthalmic diseases, like age-related macular degeneration and diabetic retinopathy.^{9,53} Beyond ophthalmology, AI has been successfully evaluated in general pathology.⁵⁴ A meta-analysis published in 2019 compared deep learning performance against health-care professionals in detecting general diseases from medical imaging,⁵⁵ showing similar performance between deep learning and health-care professionals.

4.2 | Binary algorithm versus the holistic role of the physician

The performance of deep learning is comparable to trained ophthalmologists for glaucoma classification based on fundus photographs in an established protocol, with an analysis time of 10% of the reading time of ophthalmologists.²⁵ Further studies should evaluate the algorithms in clinical practice, with larger sample size and different datasets. Although fundus photographs are an accepted modality for screening, it is well established that subjective evaluation of optic disc suffers from low reproducibility from ophthalmologists,^{56,57} whereas in one study,²⁵ the deep learning system Pegasus showed perfect repeatability. It is difficult for deep learning and humans to classify glaucoma in eyes with less severe manifestations or multiple comorbid eye conditions like myopia.⁵⁸ The diagnosis of glaucoma relies on a compilation of risk factors and functional testing, not only on structural testing of the optic nerve. In all our studies, the algorithms and the ophthalmologists are evaluated on a binary classification of referable glaucomatous optic neuropathy, not on the diagnosis of glaucoma itself. The algorithms should be evaluated after integration of other clinical parameters (intra-ocular pressure, central corneal thickness, ...). Until additional investigations, it is unlikely that a binary deep learning system will be able to replace the holistic role of an ophthalmologist.

4.3 | Deep learning, glaucoma screening, and tele-medicine

Screening for glaucoma is essential given its long asymptomatic nature.⁵⁹ Thus, early detection of glaucomatous changes is important and may be improved by advanced technologies. Considering that screening for glaucoma is costly, time-consuming, and limited by the experience of the ophthalmologist⁶⁰ with limited reproducibility,⁶¹ advanced tools like deep learning, internet of things (IoT) and 5G⁶² may play a pivotal role in the field of glaucoma, especially in low-income countries usually associated with

a low density of ophthalmologists.⁷ One advanced technology already used for glaucoma screening is tele-medicine, which effectively detect glaucoma in patients^{63,64} and can prevent functional loss caused by extensive delay in ophthalmic care.⁶⁵ AI can complement tele-medicine by adding an automated part by recognising effectively glaucomatous changes in fundus images, optical coherence tomography scans and visual fields. The COVID-19 pandemic recently provided an ecosystem in which health-care systems had to rapidly develop new ophthalmic care delivery by combining IoT, 5G and AI. Indeed, ophthalmologists are an at high-risk profession to SARS-CoV-2 spreading.⁶⁶ Thus, telemedicine associated with AI could mitigate the risk of virus propagation, allowing healthcare access for the patients as well as respect of sanitary precautions.⁶⁷ In a context of ageing of the population and COVID-19 pandemic, ophthalmologists could benefit from these innovations to relieve the consultation pressure on medical institutions.⁶⁸ Deep learning could change the practices in glaucoma management, allowing diffusion of tele-medicine in the field of ophthalmology, especially in a pandemic context like COVID-19, in rural and low-income countries with poor medical access.

4.4 | Limitations

We inherited the limitations of all meta-analyses,⁶⁹ as well as the limitations and bias of the individual studies of which they are composed. Furthermore, we conducted the meta-analyses on only published articles, so they are theoretically exposed to publication bias. Despite the meta-analysis is based on a moderate number of studies, the use of broader keywords in the search strategy limits the number of missing studies. In addition, some studies were monocentric, limiting the generalizability of our results. Not all studies reported the characteristics of the population studied. Included studies were homogeneous according to funnel plots, but the populations were not equally distributed around the world with a majority of Caucasians in glaucoma and control groups. The age of our patients was quite representative of the majority of glaucoma patients but the prevalence of glaucoma is not representative of the world population and precludes generalisation. The studies included were not clinical trials, precluding robust conclusions for our meta-analyses. Though there were similarities between the inclusion criteria, the population, glaucoma definition, and deep learning used were not identical. Although AUC is a commonly-reported metric to assess deep learning performance, it lacks clinical interpretability because it measures performance across all thresholds, including both clinically relevant and irrelevant ones. Also AUC treats

sensitivity and specificity as equally important when averaged across all thresholds, but the clinical consequences of sensitivity are far more important for a screening test. We retrieved a limited number of studies in our meta-analysis, which is explained by the quite recent development of deep learning and their use in medical imaging. Most importantly, we believe that the key message of our article is not altered.

4.5 | Conclusion

We demonstrated a similar performance of deep learning compared to ophthalmologists in glaucoma diagnosis from fundus examinations. These results could have great implications in glaucoma diagnosis and to improve the global eye health. Tele-medicine assisted by AI applied on images in ophthalmology field could upset the clinical practices, especially in a pandemic context like COVID-19 or in low-income countries with poor medical access. Further studies should evaluate the glaucoma diagnosis performance of deep learning in clinical conditions.

CONFLICT OF INTEREST

The author declares there is no potential conflict of interest.

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

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